



UNIVERSITY
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Eurostar Customer Satisfaction Analysis
Bachelor's Thesis Business Analytics

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Introduction

The recent merger between Eurostar and Thalys has fundamentally reshaped the landscape of European high-speed rail. The recent merger, which occurred in October 2023, has cemented Eurostar in a key position as a transportation provider within Europe by creating a unified network between the channel within the United Kingdom and mainland Europe. We aim to utilise the diverse range of data that Eurostar collects on certain KPIs and relevant metrics to be able to improve both customer satisfaction and the Net Promoter Score (NPS) within Eurostar. There is currently a missing link between specific operational data, such as rolling stock maintenance, station throughput, and train regularity, and the data received from customer feedback. Our role is to be able to bridge this analytical gap. This project is motivated by the need to move beyond descriptive survey results toward a predictive, data-driven understanding of customer satisfaction and informed decision-making.

Business Relevance

The business relevance of this study is grounded in operational excellence and improving customer satisfaction, which aligns with business objectives. For Eurostar, by improving the customer experience and NPS score on an overall basis, not only provides a better reputation for the business but also provides a clear financial impact as customers recommend the service to others, making Eurostar a more competitive player within the transportation sector. By having our group analyse and establish the link between certain operational factors and the resulting NPS, it provides Eurostar with areas they can act to improve upon and focus future budgets to optimise their operational efficiency. Furthermore, understanding the different trade-offs, such as the service levels and price, can bring new ways for the company to gain insight into the behaviour of the consumers.

Research Questions

The research is centred on the critical transition from a descriptive evaluation of service quality to a more advanced, predictive framework that informs strategic decision-making. Eurostar currently faces a complex data challenge that, if effectively analysed, will put it in a powerful and crucial position. While there exists detailed information about internal operational performance, these technical data streams frequently stay apart from the qualitative insights obtained through Net Promoter Scores (NPS) and customer satisfaction surveys. In order to pinpoint the precise operational drivers that most strongly influence passenger sentiment, this research attempts to unify these diverse statistics. All recorded factors influencing a customer's end-to-end travel experience from the check-in stage through to arrival at the destination will be analysed using various statistical and logical models. In the end, this context lays the groundwork for a data-driven strategy that aims to measure the quantifiable influence of both controlled and uncontrolled operational factors on sustained brand loyalty. The impact analysis of the variables obtained from the research will serve as a critical investment guide for Eurostar management.

Primary Research Question: To what extent do individual operational performance indicators and ticket pricing influence Eurostar's Net Promoter Score (NPS) and customer satisfaction, and how can these relationships be modelled to prioritise operational improvements that maximise customer satisfaction?

Subquestion:

At what individual operational threshold levels, and through which combinations of threshold breaches, do journey-stage factors produce disproportionate declines in the NPS, and how can these effects inform operational prioritisation?

Eurostar already knows, in broad terms, that its operations shape how passengers feel about the service. What it does not know is where the damage starts. A five-minute delay and a forty-minute one are not the same thing, and the relationship between a delay and a passenger's reaction is unlikely to be a straight line. At some point along each operational factor's scale, the recommendation score stops slipping gradually and begins to fall sharply. This subquestion looks for those points. It asks two things: at what threshold individual journey-stage factor begins to produce a disproportionate fall in the recommendation score, and hence in NPS; and which threshold breaches, when they occur together, do more harm than their separate effects would predict. Both questions feed the same practical end. If Eurostar can see which breaches cost the most NPS per occurrence, and which combinations turn a manageable problem into a poor journey overall, it can direct limited operational budget at the failures that matter most rather than spreading it evenly across all of them.

Artifact requirements and goals

The artifact exists to help the operations teams & managers at Eurostar to improve the overall efficiency and effectiveness of their rail operations and prioritise the factors with the most impact in relation to price. This can be achieved through the artifact providing information about the relationships between different operational indicators and customer satisfaction and NPS, with rankings in terms of impact and urgency. Current KPIs only measure the effectiveness of existing processes from an operational and logistics viewpoint, and fail to take customer satisfaction and NPS into consideration as meaningful metrics for business decision-making.

Description of the final artifact to be delivered

The final artifact should provide an in-depth analysis of operational factors' effect on customer satisfaction and NPS, along with a ranking by impact and priorities. This analysis is to be supplemented with visualisation deliverables. Included in the analysis are findings on how these relationships vary with price, and scenario analysis for customer sentiment in case of unexpected situations. The artifact will help Eurostar decide on which operational domain to prioritise and determine the resources to invest in each factor proportional to their impact. Furthermore, it should give an overview of how the system works and how each part in the system affects the others using Directed Acyclic Graphs and journey simulations. The final form of the artifact will be determined after data access has been given and further communication with the company, and will closely reflect the organization's needs, with dashboards being the current candidate.

Background

Description of the method/methodology used

This research adopts a hybrid methodology that balances scientific rigour with business relevance to develop a predictive analytics model (the artifact) for Eurostar.

Design Science Research (DSR)

Design Science Research is the overall framework guiding this project. It is a research paradigm focused on creating and evaluating artifacts, such as models, methods or prototypes, that solve real-world problems while contributing to theoretical knowledge, and it complements business analytics work in which the deliverable is a designed solution whose effectiveness must also be validated (Hevner, 2007; Zwetsloot, 2025). Its central idea is the balance between rigour, the grounding of the work in scientific theory and peer-reviewed literature, and relevance, the engagement with practical problems and stakeholders, such that the artifact is actually applicable (Hevner, 2007). For this project, DSR provides the structure within which a functional artifact can be built to fill the analytical gap between Eurostar's operational data and the customer-perception measures the business already collects. We employ the three-cycle model.

Relevance Cycle: By establishing precise needs through stakeholder collaboration with data engineering and logistics teams, we connect our study with the Eurostar's operating environment.

Design Cycle: The main iterative procedure where we construct the Databricks prediction model and assess its effectiveness using previous NPS data.

Rigor Cycle: Reviewing scholarly references on service-quality models and advanced analytics, such as multi-factor regression and graph theory, would help to ground the project in the knowledge base.

CRISP-DM (appendix)

While DSR focuses on relevance and iteration at a high level, CRISP-DM provides the operational backbone through its six phases: business understanding, data understanding, data preparation, modelling, evaluation, and deployment, giving the project a concrete procedural structure (Chapman et al., 1999). Its advantages benefit the project with its **iterative refinement**. The phase sequence is explicitly non-linear (Chapman et al., 1999); inadequate model results send the project back to data preparation, and an evaluation revealing the wrong KPIs sends it back to business understanding. The methodology assumes this will happen, which matters given how complex integrating operational and survey data across a multinational rail network is. Moreover, the phased structure keeps each operational KPI ("X") explicitly tied to NPS ("Y"), reinforced through DAPS diagrams, so operational factors do not get disconnected from the outcomes they are meant to explain across datasets of different types and sources, hence **maintaining consistency**. (Chapman et al., 1999). Finally, MoSCoW, applied within this structure, **anchors the project** to a Minimum Viable Artifact addressing Eurostar's core requirements, which matters given how many directions the analysis could otherwise expand into (Martínez-Plumed et al., 2021). A structured, documented process also gives the team and Eurostar confidence that conclusions aren't just the product of whichever model ran first (Mendes & Farinha, 2026). DSR and CRISP-DM are not interchangeable: DSR establishes what is being built and why it matters to Eurostar, while CRISP-DM

defines how the work is carried out, and the project stays systematic precisely because both are in play (Martínez-Plumed et al., 2021).

Business Understanding

Reviewing Domain and Artifact Literature

The Net Promoter Score (NPS), introduced by Reichheld (2003) and popularised through the Harvard Business Review, rests on a single "likely to recommend" question: scores of 0-6 are "detractors", 7-8 "passively satisfied", 9-10 "promoters", and NPS is the percentage of promoters minus the percentage of detractors (Grisaffe, 2016). Reichheld claimed this one number was sufficient on its own to manage customer loyalty, a claim later scholarship has not supported.

Fisher and Kordupleski (2019) argue NPS gives no statistically reliable guidance for retaining or winning customers, only a snapshot of organisational performance, and Keiningham et al. (2008) similarly question its superiority over alternatives such as the Norwegian Customer Satisfaction Barometer. Both see more value in NPS as one input within a broader system, such as Bain's own Net Promoter System, which pairs the score with follow-up questions on why customers feel as they do. Korneta (2018) supports this empirically: within Polish transportation companies, NPS included in a metric portfolio had a significant positive relationship with profitability (ROA, ROS, ROE).

What drives the score matters more here than the metric debate itself. Allen et al. (2020) find critical incidents are the strongest driver of transit dissatisfaction, that operator response to an incident matters as much as the incident itself, and that waiting time is weighted more heavily than travel time. Berggren et al. (2026) show cleanliness spills over into how punctuality is perceived, so domains are not judged independently; Cantwell et al. (2009) link crowding to commuting stress through lost personal space and control; and Van Lierop et al. (2018) point to personalising the journey, through mode, fare and schedule, as the more durable lever for satisfaction.

Together, this literature shows operational factors interact rather than act independently, and their effect on satisfaction is unlikely to be linear, which is the gap this project's threshold and combination analysis is built to close.

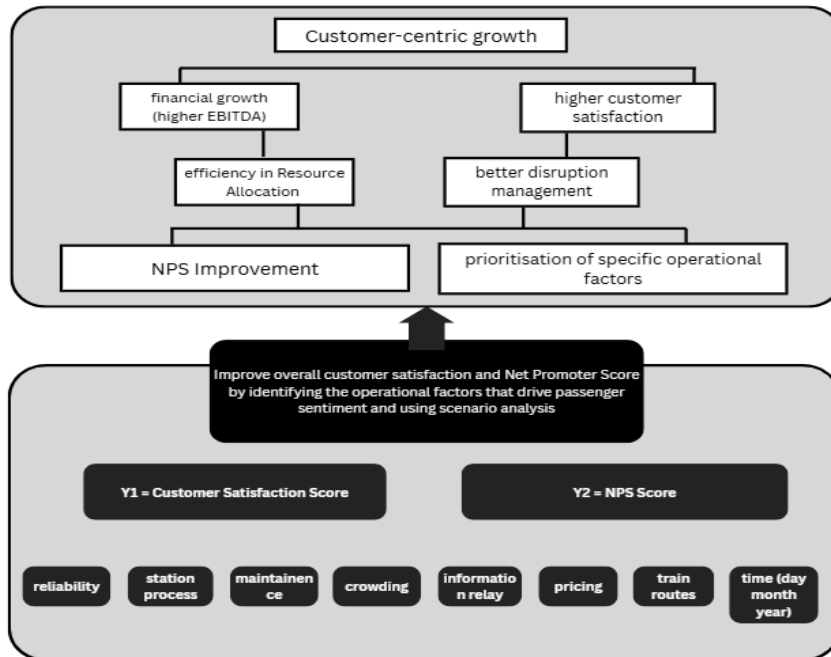


Figure 1. DAPS diagram

Business Value

Eurostar's growth strategy is built around becoming more customer-centric. As **Figure 1** shows, the idea is that improving how passengers experience the service is not separate from financial performance but directly connected to it. Better disruption management leads to higher satisfaction, which feeds NPS improvement, which in turn supports both revenue growth and more efficient allocation of operational resources.

What makes this project relevant to that strategy is the gap that currently sits in the middle of it. Eurostar collects operational data (reliability, station processes, maintenance, crowding, information relay, pricing, train routes, time) and separately collects customer feedback through satisfaction scores and NPS. Both exist, but the link between them is not quantified. The business broadly knows that operations affect customer sentiment. However, It does not know which operational factors matter most, at what levels they start to lower NPS and customer satisfaction, or which combinations make things disproportionately worse. That means the prioritisation of operational factors, the business needs to direct resources efficiently, is currently done without a data-driven basis for deciding what to fix first.

The two dependent variables, Customer Satisfaction Score and NPS, are chosen because they are the metrics Eurostar already uses to measure whether the customer-centric growth strategy is working. Making the relationship between operational inputs and those outputs explicit is what closes the gap the business has identified and gives the prioritisation of factors the quantitative backing it currently lacks.

Design principles

- Interpretability over complexity and performance (the artifact focuses on interpretations rather than complicated or black-box models, just for performance)
- Modularity (the artifact supports individual analyses (sub-questions) alongside the main research question)
- Causal interpretation (the artifact distinguishes correlation from causation and does not make any causal claims without substantial proofs and rigorous testing)
- Reproducibility (the analysis pipeline is fully documented and reproducible)

MoSCoW

MoSCoW is a prioritisation framework used to categorise the requirements of a project. In a design-science project, it balances rigour and relevance, since the *Must have* tier forces the artifact to address the real-world problem rather than drift into work that is technically interesting but business-irrelevant. It also manages scope by preventing feature creep and committing the project to a minimum viable artifact, and provides a shared language for stakeholders, advisors and team members to discuss what the artifact will and will not deliver (Zwetsloot, 2025). For such a project as ours, where analytical directions could expand indefinitely into adjacent areas such as real-time monitoring, that discipline matters.

The MoSCoW table, as a summary, applies the framework to the Eurostar artifact. “*Must have*” requirements capture the core analytical commitment: quantifying and comparing the effects of operational factors on customer satisfaction and NPS, examining how price moderates them, and analysing each component in detail. “*Should have*” requirements extend the work into actionable outputs, namely baseline targets per travel tier and the analysis of operational clusters, which strengthen the artifact without being strictly required for it to function. “*Could have*” requirements hold the station-network and end-to-end simulation work, which would add depth, but depends on data access and time. “*Won't have*” requirements rule out real-time predictions, demand forecasting and individual-level behaviour modelling, all of which fall outside the system-level scope of the research question. Categorising each requirement in this way keeps the analytical pipeline anchored to what Eurostar actually needs.

Data Understanding

Overview

Passengers are surveyed online via Qualtrics one working day after their journey regarding overall satisfaction, journey-related opinions, and likelihood to recommend Eurostar. Satisfaction items use a five-point Likert scale (1 = very dissatisfied/strongly disagree, 5 = very satisfied/strongly agree); recommendation likelihood runs 0 (not at all likely) to 10 (extremely likely).

The questionnaire has two steps. A main block, shown to everyone, covers comfort, cleanliness, staff manners, punctuality, information provided, departure-station experience, ticket-price satisfaction, overall rating and the recommendation score. One in four respondents is then randomly assigned a touchpoint

block, a more granular, journey-stage-specific version of the same content (toilet cleanliness, noise, atmosphere, staff helpfulness at that particular stage, for example). These touchpoints are used internally by Eurostar as aspects of the journey in which customers interact directly with services delivered by the company

The touchpoints retained for analysis are onboard experience, the departure-arrival process, and staff interactions. Several others, disruptions, onboard cleaning impact, catering, lounge and Eurostar Café, only appear when a specific event occurs or service class allows it, and average 85% missing values; the lounge, catering and café ones are dropped, while disruption and cleaning-impact touchpoints are kept given their direct link to operational performance.

The survey also captures age, residence, language, and travel reason for segmentation, plus journey metadata: train number, origin-destination, date, seat and coach, and connection status, with train number and date as the key linking responses to operational data.

Hypothesis

H1 - Arrival delay has a non-linear, threshold-structured effect on NPS

H2 - Compensation moderates the NPS impact of delay among passengers delayed beyond 60 minutes

H3 - Early journey delay has an independent negative effect on NPS, separate from arrival delay

H4 - Delay caused by controllable, Eurostar-attributable failures costs more NPS per minute than externally caused delay

H5 - Routine cleaning score buffers the NPS impact of delay: high-scoring services absorb delay better than low-scoring ones

H7 - When both arrival delay and early journey delay are elevated simultaneously, the combined NPS penalty exceeds the sum of their individual effects (dual-delay compounding)

H8 - Fleet age (TGH vs. RUB) affects NPS independently of delay, and the effect is asymmetric between Detractor and Promoter outcomes

H9 - Ticket price has an NPS effect of its own, distinct from perceived price satisfaction

H10 - High train occupancy (above ~400 passengers) is associated with lower NPS, consistent with crowding effects

Data Filtering

There are 299005 survey responses in total, consisting of journeys departing from April 30th 2024 to May 13th 2026. Before further analysis, the raw survey responses are filtered based on target variables, linking keys, and Qualtrics built-in data quality indicators: duration, straight-lining, and unanswered questions. The decision rules and resulting exclusions are summarised below.

a. Survey duration

Duration was measured in seconds from survey start to submission. The distribution is heavily right-skewed, with a mean of 46 minutes and a median of 6.4 minutes. This large difference between the

mean and median value can be attributed to the presence of extreme outliers: the maximum duration is 13 days, and the 99th percentile is 14 hours. Therefore, the median duration of 6.4 minutes will be used as a benchmark instead of the mean. Most people take between 2 and 10 minutes to answer the survey when examining the distribution; therefore, responses completed in under 2 minutes (0.3% of the sample) were deemed too fast, while those exceeding one hour were also interpreted as signs of disengagement, which were then excluded.

a. Straight-lining

Straight-liners are respondents who give the same option across multiple questions to quickly get through the survey without reading. Qualtrics algorithm marks a response when straight-lining occurs in at least 3 question-matrix tables, or in 80% of the question-matrix tables. The median and mean values were 25% and 32%, respectively, which is not informative on their own. To determine whether straight-lining genuinely indicated low-quality responses, the distribution of recommendation scores and survey duration was also examined per straight-lining quartile. Passengers in the top quartile showed the highest mean recommendation score and the shortest median duration. This could potentially skew and cause bias in the data, and since this portion takes up only 4.5% of total observations, it was removed accordingly.

b. Unanswered questions

A response is flagged when 30% or more of the questions shown are left unanswered. On average, 4% of the questions were skipped, with the 95th percentile reaching 13% and the maximum being 21%. Customers with unanswered questions in the range of 15% - 21% also had the lowest recommendation scores. This could be due to genuine dissatisfaction, disinterest, or non-engagement. While the cause is hard to verify, since these observations only take up 0.3% of the sample, they were dropped from the dataset.

Variables Preparation

After filtering based on Qualtrics indicators was applied, any remaining observations with missing values for overall rating and recommendations were discarded as they served as mediators and target variables, respectively. Responses lacking a train service number were also excluded because they cannot be reliably linked to operational factors.

Since the answers were given in text format, they were applied ordinal encoding to preserve the original Likert 1-5 integer scale. The recommendation scores were left in their original form of 0-10, though ordinal encoding from 0 to 2 was used on the NPS category assignment column of “Detractor, Passive, Promoter” respectively. The conditional blocks (catering, lounge, café) were removed entirely due to their disproportionate missing values and inability to be assigned to operational KPIs. Questions that have more than 80% missing values within a touchpoint were also dropped

Literature review: Relevant variables/ Data processing

The variables Eurostar collects (reliability, maintenance, crowding, information relay, pricing, survey sentiment) map onto what earlier studies have built, and those studies are most useful for showing how a raw operational feed becomes information a model can use. Monsuur et al. (2021) are the closest comparison: instead of perceived delay, they used Network Rail's objective records (real-time delay

minutes, scheduled travel time, service frequency, cancellation and fleet-failure flags) and split delay into six cause groups (fleet, external, train operator, network management, track, operations), separating controllable from uncontrollable disruption, the same line Eurostar needs to draw.

Which variables make it into a model usually comes down to what the linked systems can supply and what an operator can act on. Ruiz et al. (2024) leaned on three operational predictors, average waiting time, occupancy and speed, all rebuilt from smart-card and GPS records rather than survey items. Verhoef et al. (2017) built variables from internal records too, found punctuality and seat availability the main drivers, and used factor analysis to avoid double-counting correlated inputs. Survey-based studies follow the same pattern: Eboli and Mazzulla (2015) sorted 33 attributes into constructs including service and comfort, and de Oña (2021) found intermodality, frequency, punctuality and speed mattered most.

Several studies also pruned variables statistically before modelling: Monsuur et al. (2025) dropped attributes correlating above 0.8 and kept only items answered by at least 90% of respondents; Verhoef et al. (2017) used stepwise selection and a Wald test; Anwer et al. (2024) used factor analysis to compress a large attribute set. This supports Eurostar's own decision to strip sparse conditional touchpoint blocks, drop high-missing questions, and exclude responses with no train service number, since without that key the operational variables cannot be attached. The target variable receives the same treatment.

Preprocessing plan & Data quality

The preprocessing stage links the cleaned 178,384-response survey file to each journey's operational conditions. Since the survey is recorded per respondent, but most operational variables (delay, maintenance, cleaning, occupancy) are recorded per train service, passengers on the same service share operational values while keeping their own scores. Work proceeded in two stages: integrating sources and building features, then auditing the result for sparsity, patterned missingness, redundancy and modelling suitability, since feature quality could only be judged once everything was joined.

Integration of operational sources. The filtered survey (178,384 rows, 211 columns) was joined to external tables on a train-number-plus-date key, using left joins throughout to preserve every response and avoid many-to-many duplication. Three joins need particular care. Arrival-cause data added cause category, domain and responsibility, but 38.6% of rows had no matched responsibility value, and raw categories were unevenly distributed (traffic regulation alone covered 57,827 rows; several causes appeared fewer than 35 times), so causes are represented at two levels, a broad seven-category grouping and a finer selective set. Critically, 11,023 of the 68,932 rows originally labelled No_Delay had a positive recorded delay with no matched cause (NPS 7.62 versus 8.49 for genuinely zero-delay rows); these were reclassified as Delayed_NoCauseRecord and the cause dummies rebuilt. Passenger and booking data, aggregated from 7.1 million raw records to service level, overlapped only 44.4% of survey-linked services; total_pax was missing for 39.1% of rows, almost entirely structurally (correlation -0.913 with Channel status, since passenger data covers Channel services almost exclusively) rather than reflecting genuine low occupancy, hence the added group_data_available indicator.

Feature construction followed the same logic throughout: inspect the raw distribution, then build representations suited to threshold detection. Arrival delay was kept continuous (mean 12.2 minutes, range -106 to 739) alongside banded indicators (37.2% of rows exceed 5 minutes, 22.9% exceed 15, 13.2% exceed 30, 5.5% exceed 60); the long tail means severe-delay thresholds will need wider

confidence ranges. The staff-service composite (visibility, friendliness, helpfulness) was empirically justified before combining, with inter-item correlations of 0.674–0.854 and a Cronbach's alpha of 0.884. Cleaning and preventive-maintenance variables were 33-50% missing, and corrective-maintenance fault counts were 66.3% zero overall, but nearly every pattern split along Channel status (correlations -0.877 to -0.98), so tracking and availability indicators were added throughout, and these findings are flagged for comparison with and without Channel observations rather than treated as network-wide by default.

Missing-data strategy. The pre-imputation file held 178,384 rows, 186 variables and 6.19 million missing cells, with missingness treated as three distinct types: structural survey rotation, source-coverage gaps, and ordinary item-level absence, rather than one blanket rule. Twenty-eight variables exceeding 75% missingness were dropped outright; the remainder were imputed by concept (cleaning and maintenance recency by equipment-type median with a `was_tracked` flag; passenger and group variables by median with a `group_data_available` flag; remaining numerical predictors by simple median), affecting 119,873 of 178,384 rows. Two Parquet files were exported: a full dataset (168 columns) and a lean dataset (154 columns), dropping the 14 weakest-coverage maintenance and cleaning fields, so later results can be checked for sensitivity to the most heavily imputed variables.

Limitations. Row-level splits can place passengers from the same service in both train and test sets, so validation needs to account for within-service dependence. Several variables' availability is patterned by Channel and equipment type rather than missing at random, so network-wide claims need checking against Channel-only subsets; the corrective-maintenance zero pattern on Channel services is most likely a coverage gap. Severe-delay thresholds carry more sampling uncertainty than central-range ones, and imputation preserves sample size at some cost to genuine variation, hence the required full/lean comparison. Throughout, results remain observational associations, not causal estimates.

Preliminary Decision-Tree Analysis

A decision-tree analysis was run as the first modelling stage, ahead of the gradient-boosted model, to check whether the prepared data contain the non-linear, conditional structure the research question expects, not to produce final operational thresholds. The recommendation outcome was modelled four ways: a regression tree on the raw 0-10 score, and three classification trees for detractor (0-6), passive (7-8) and promoter (9-10) status, since the point where Eurostar starts losing promoters need not coincide with the point where passengers become detractors.

Six feature specifications were compared to separate objective operational signal from subjective evaluation: the full set (operational, satisfaction, profile, price, staff and temporal variables together); satisfaction excluded; satisfaction and profile both excluded; operational variables alone; operational plus metadata price (labelled `just_price` in the notebook, despite the name); and operational plus price plus the staff composite (`price_and_staff`). The operational-only and price/staff specifications are the ones most directly aligned with levers Eurostar can act on; the broader sets serve as a ceiling check.

The 178,384 rows were split 80/20 by row, not by service, with the same split held fixed across specifications. Tree depths 4 through 8 were tested, with a 1,000-observation minimum leaf size to avoid thresholds supported by small or unusual groups. Regression was scored by test R-squared and RMSE, classification by ROC-AUC with balanced class weights given the unequal group sizes. The row-level

split is a known limitation: passengers from the same service can land in both train and test, which likely flatters fit relative to genuinely unseen services, so the GBM stage moves to service-grouped validation.

Results

Predictive fit tracked how subjective the inputs were. The full set reached a test R-squared of 0.488 at depth 8; excluding satisfaction dropped this to 0.268, and excluding profile and demographic information too, to 0.250. The operational-only tree reached only about 0.111, flat past depth 5-6; adding price moved this to about 0.118; adding price and the staff composite together moved it to about 0.250, meaning staff experience, not price, does almost all of that work. Classification followed the same ordering everywhere: detractors easiest to separate (full-feature ROC-AUC about 0.90), promoters next (about 0.83), passive respondents hardest (about 0.70, falling to near-random at about 0.53 with operational variables alone). Operational data says far more about strongly negative experiences than about the ambiguous middle category.

Arrival delay was the most consistent operational variable, splitting repeatedly across specifications in five recurring regions: 10-14, 19-21, 25-32, 37-47 and roughly 90-92 minutes. Promoter-tree splits cluster toward the lower end and detractor-tree splits toward the upper end, suggesting promoter loss may begin before detractor risk rises sharply, worth carrying into SHAP rather than assuming one threshold serves both groups. Early-journey delay split in parallel but not identical regions (roughly 3-5, 16-22, 26-42 minutes), worth testing separately rather than as a proxy for arrival delay. The staff composite produced four recurring regions (roughly 2.4-2.8, 3.4-3.8, 4.2-4.4, 4.6-4.8) and, when added, consistently split before delay and shifted the delay threshold within each branch, the clearest interaction candidate from this stage. Metadata price split only conditionally (around 59, 88-95, 160) and is confounded with class and route; ticket-price satisfaction split cleanly at 1.5 and 2.5 but is subjective, not an operational lever.

These are candidate regions, not confirmed thresholds: a single tree is unstable to resampling and gives no confidence interval. They set the search space for the GBM/SHAP dependence and interaction analysis, with arrival delay $\times$ staff composite the top interaction to test, ahead of grouped resampling to attach a confidence range to any threshold before it reaches the ordered logit or the goal-programming handoff.

Data processing pipeline

The pipeline runs as ten connected stages rather than one cleaning operation.

Stage 1: Load the cleaned, filtered survey export, with target variables and linking fields present but operational sources not yet joined.

Stage 2: Audit existing variables for dimensions, types, implausible values and missingness. Thirty-three columns were dropped before any operational data were added, reducing the file from 211 to 178 columns.

Stage 3: Prepare each operational source separately, standardising service identifiers, removing duplicates, and aggregating passenger-level records to service level, so that joins do not change the unit of analysis.

Stage 4: Join arrival-cause, passenger, cleaning and other maintenance sources via left joins, checking the row count against the original 178,384 after every merge. The stage finishes at 269 columns and the same row count.

Stage 5: Construct analytical features. Continuous measures are retained where threshold discovery needs the original scale, alongside grouped and binary versions for exploratory use. Delay causes are represented at both broad and selective levels, and passenger, staff, cleaning, maintenance, price, class, route and time variables are built according to their analytical role.

Stage 6: Run internal consistency checks, including cross-checking the No_Delay category against positive arrival-delay values (the source of the Delayed_NoCauseRecord correction), rebuilding affected dummies, confirming cause indicators are mutually exclusive, and validating the staff composite via correlation and Cronbach's alpha before combining.

Stage 7: Profile missingness as a pattern rather than just a rate, comparing it across Channel status and equipment type to judge how much absence reflects source-system coverage rather than random loss.

Stage 8: Drop variables above 75% missingness, then impute the remainder by concept-specific group with tracking and availability indicators, applying a final simple median pass only to what is left.

Stage 9: Encode model variables. Equipment type is dummy-encoded with E300 as the reference category. For interpretation, within-route-type comparisons are used: RUB vs TGH for Continental services and E320 vs E300 for Channel services, since comparing Continental fleet types against the Channel-only reference crosses the route-type divide already captured by is_channel. Exploratory duplicates such as passenger quartiles are excluded from the model specification, and non-ordered numeric codes are converted or removed.

Stage 10: Export full (168-column) and lean (154-column) Parquet files, both at 178,384 rows with no remaining missing values, as the two robustness specifications used going forward.

The pipeline resolves most technical inconsistencies but carries forward limitations that condition how its outputs should be used. Operational predictors are shared within service while the outcome is recorded at passenger level, favouring service-grouped validation over a row-level split. Passenger, cleaning and maintenance coverage is structurally tied to Channel status and equipment type, so findings from one part of the network should not be generalised without checking. Corrective-maintenance zeros on Channel services most likely reflect a recording-system gap rather than confirmed zero faults. Severe-delay thresholds rest on fewer observations and need wider confidence ranges. Imputation preserves sample size but can smooth genuine relationships, which is why the full/lean comparison is treated as a required check rather than an optional one.

Modelling & Design

Review literature for appropriate modelling choices (rigor cycle)

This subquestion has two halves: where, along each operational factor's scale, the decline in NPS begins to accelerate, and which breaches in combination prove more damaging than their separate effects

suggest. Neither relationship has a form known in advance, which rules out a single regression specified up front. The analysis is therefore staged. A flexible model first establishes the importance and likely shape of each factor, and an interpretable model then quantifies the effects. Li et al. (2020) followed this sequence in prostate-cancer survival, training a gradient-boosted tree model and generating SHAP and SHAP-interaction values before confirming a significant interaction in a Cox model. The present pipeline keeps that structure but starts with a single decision tree, which gives an early read on the direction of effects and an interim result to discuss with Eurostar, against which the analysis can be reaffirmed or redirected.

The case for a tree-based model over conventional regression is well supported. Gao et al. (2021) compared machine-learning algorithms against logistic regression on airline-satisfaction data, found Random Forest superior, and showed that service attributes affect satisfaction in piecewise nonlinear ways, with genuine interaction effects between factors. That is the structure this subquestion is built around. Alanazi et al. (2024) compared six algorithms on roughly 7,500 Skytrax reviews and found that Gradient Boosting achieved the highest mean accuracy at 88.15%, supporting the broader suitability of boosting methods for recommendation modelling; more specifically, Joung and Kim (2023) found that XGBoost produced the highest F1-score among the six machine-learning approaches they compared when predicting customer ratings from product-feature sentiments. XGBoost is therefore the core model, with the decision tree retained as a preliminary and more transparent check.

This pipeline interprets the model through SHAP, following Li et al. (2020) and the practical guidance of Ponce-Bobadilla et al. (2024). Dependence plots relate a factor's value to its contribution to the prediction, so a sharp change at a candidate point can be tested. Li et al. (2020) used this to assess a clinical cut-point and found the relationship was close to linear, which is a reminder that the method rejects thresholds as readily as it confirms them. Interaction values strip out both main effects to isolate what a pair of factors does jointly, yielding a ranked shortlist for examination. The reason for using SHAP rather than the alternatives is principled: Lundberg and Lee (2017) prove that within the class of additive feature-attribution methods, SHAP is the only one satisfying local accuracy, missingness and consistency simultaneously, the last of which is the property that lets a ranking of operational drivers be trusted as a basis for prioritisation.

The thresholds and interactions that emerge are then encoded as explicit variables and fitted with an ordered logit, since NPS is ordinal. Monsuur et al. (2021) are the closest precedent. From roughly 7,000 UK National Rail Passenger Survey responses joined to Network Rail records, their ordered logit found passengers reacting negatively beyond 30 minutes of delay and recommended prioritising trains nearing the 10-minute mark. Their target was a five-point satisfaction scale rather than NPS, but the logic carries over directly. The outcome for Eurostar is a ranked, quantified account of which breaches cost the most NPS per occurrence, which is the prioritisation basis the business does not yet have.

Develop core model/algorithm (design cycle)

XGBoost

The core model was implemented using XGBoost to capture non-linear relationships and interactions between operational factors and passenger recommendation. For each feature set, two XGBoost models

were trained in parallel: a regressor predicting the raw 0–10 NPS score and a classifier predicting NPS category (Detractor, Passive, Promoter). Running both on identical data allows regression R^2 and classification accuracy and AUC to be compared directly, which is useful given the ordinal nature of the target variable. The regression specification preserves more detail across the recommendation scale, while the classification model reflects the established business categories. Since the recommendation score is ordinal, treating it numerically is used only as a practical modelling approximation; the final thresholds will later be examined using ordered logit.

Six feature sets were constructed as a nested staircase, each adding one block of variables to the previous:

Feature set	Description
operational_only	Delay variables, equipment, cleaning, maintenance, faults, disruption flags
just_price	Operational + raw ticket price (metadata_price)
price_and_staff	Operational + price + staff composite
profile_excluded	Adds passenger profile (travel reason, booking channel, loyalty)
satisfaction_excluded	Adds in-journey perception items excluding NPS-adjacent ratings
everything	Full feature set including all satisfaction items

Each step in the staircase isolates the marginal contribution of one additional block, so the R^2 gap between any two adjacent sets can be attributed to the variables added at that step. The two sets of primary interest for the research question are operational_only (estimates total operational effects) and price_and_staff. The remaining sets serve as diagnostics.

Validation was performed at train-service level. A service group combined train number and travel date, ensuring that passengers from the same journey were not split across training and test folds. The goal was to have a more realistic estimate of performance on unseen services and avoid overstating model accuracy through repeated operational conditions. All experiments used GroupShuffleSplit cross-validation, grouping on journey date to prevent data leakage across time periods. Each configuration was evaluated over five folds. The final model was trained on 151,272 observations and evaluated on a held-out test set of 27,112 observations, representing 15% of the full $N = 178,384$.

Hyperparameter search ran in two stages. Stage 1 tested four configurations across all six feature sets, varying tree depth (2–5) and learning rate (0.03–0.05), with subsample = 0.8, colsample_bytree = 0.8, and early stopping on validation loss. Stage 2 narrowed the search to the two focal feature sets and refined the configuration around the Stage 1 optimum. The selected final configurations were:

- operational_only: max_depth = 4, learning_rate = 0.05, min_child_weight = 10; 235 trees (regressor), 203 trees (classifier)

- price_and_staff: max_depth = 2, learning_rate = 0.05, min_child_weight = 15; 554 trees (regressor), 650 trees (classifier)

Hyperparameter search ran in two stages. Stage 1 tested four configurations across all six feature sets, varying tree depth (2–5) and learning rate (0.03–0.05), with subsample = 0.8, colsample_bytree = 0.8, and early stopping on validation loss. Stage 2 narrowed the search to the two focal feature sets and refined the configuration around the Stage 1 optimum. The selected final configurations were:

Results

Feature set	R ²	Accuracy	AUC (Detractor)	AUC (Promoter)
operational_only	0.116	54.2%	0.681	0.626
price_and_staff	0.257	57.2%	0.796	0.727

Because GroupShuffleSplit randomly partitions the data, single-split results carry meaningful variance. A ten-seed robustness check (seeds 101–110) confirmed that the holdout figures sit within the expected range:

- operational_only: $R^2 = 0.120 \pm 0.011$, accuracy = $53.8\% \pm 0.3$ pp
- price_and_staff: $R^2 = 0.272 \pm 0.009$, accuracy = $57.0\% \pm 0.3$ pp

The gap between the two focal feature sets, roughly 15 percentage points of explained variance and 3 percentage points of classification accuracy, captures the combined contribution of ticket price and staff perception. Within the full staircase, price_and_staff already recovers 44% of the variance explained by the everything model ($R^2 = 0.599$) despite containing only 114 of 159 features. The remaining gap between price_and_staff and everything is attributable almost entirely to in-journey satisfaction ratings.

The regressor and classifier do not tell the same story. Detractor AUC for price_and_staff (0.796) is notably higher than Promoter AUC (0.727), consistent with the ordered logit findings that show sharper marginal effects at the lower end of the NPS scale. Operationally, separating those who will give a score of 0–6 turns out to be considerably more tractable than identifying those who will give a 9 or 10. This asymmetry runs through all modelling approaches in this thesis.

XGBoost is therefore used as a discovery model rather than as the final decision rule. Its role is to learn flexible patterns without imposing the candidate thresholds identified by the preliminary trees. This follows the staged use of boosted trees and SHAP demonstrated by Li et al. (2020), while Joung and Kim (2023) provide a direct precedent for interpreting XGBoost with TreeSHAP. The preferred model will subsequently be examined using SHAP dependence and interaction values to identify stable threshold regions and combinations of operational failures. These results will then be checked across the full and lean datasets and quantified using ordered logit before being used for Eurostar’s operational prioritisation.

SHAP

SHAP (SHapley Additive exPlanations) values were computed using TreeExplainer on the final fitted XGBoost models. Three types of output were extracted: mean absolute SHAP values for global feature importance (both regressor and classifier), per-class SHAP importances from the classifier (Detractor,

Passive, and Promoter separately), and pairwise SHAP interaction values identifying feature pairs whose joint effect exceeds the sum of their individual contributions.

All results are reported for both `operational_only` and `price_and_staff`. The two feature sets are treated as complementary rather than competing: `operational_only` estimates total operational effects without confounding from staff perception or price, while `price_and_staff` shows how those effects shift once the two strongest non-operational predictors are included.

Feature Importance

Arrival delay is the dominant predictor in both feature sets. In the `operational_only` regressor, `delay_gt30` alone accounts for 37% of total feature importance; the four delay threshold dummies and the continuous `arrival_delay_minute` together account for 71%. In `price_and_staff`, `delay_gt30` remains the top predictor (30%), but `staff_composite` enters at second place (10%), followed by `delay_gt60` (8%) and `arrival_delay_minute` (7%).

Two secondary operational variables appear consistently in the top ten across both feature sets: `equip_TGH`, the older Continental trainset, and `cm_open_wifi_resid`, the residualised WiFi fault rate after partialling out route effects. `early_journey_delay_minute` also ranks in the top ten for both classifiers but is more prominent there than in the regressors, which is consistent with early journey delay acting more as a threshold trigger than a continuous linear driver.

Per-Class SHAP Importances

Global feature importance treats NPS as a symmetric scale. The per-class breakdown separates which features push passengers toward Detractor from those that push them toward Promoter, and the two pictures look quite different.

Top features by class and feature set (mean |SHAP|)

Operational only - Detractor: `arrival_delay_minute` (0.168), `early_journey_delay_minute` (0.092), `had_disruption` (0.058). The three largest predictors of floor-side NPS are all delay-related.

Operational only - Promoter: `arrival_delay_minute` (0.128), `early_journey_delay_minute` (0.061), `equip_TGH` (0.048). Delay still leads, but `equip_TGH` appears in the Promoter top three while it does not feature in the Detractor top eight at all.

Operational only - Passive: magnitudes are much smaller across the board. `equip_TGH` is the top predictor (0.020), which points to fleet type primarily influencing whether passengers convert from Passive to Promoter rather than from Passive to Detractor.

Price and staff - Detractor: `staff_composite` (0.537), `metadata_price` (0.171), `arrival_delay_minute` (0.169). Staff perception dominates floor-side NPS by a wide margin once it is included.

Price and staff — Promoter: `staff_composite` (0.375), `arrival_delay_minute` (0.134), `metadata_price` (0.101), `equip_TGH` (0.064). Staff and delay are similarly weighted on the ceiling side.

The `equip_TGH` pattern needed further analysis since the raw numbers appear contradictory at first glance. The RUB trainset has both higher mean NPS (8.02 vs TGH's 7.73) and a larger ordered logit

coefficient relative to the E300 reference (0.366, significant, vs TGH's 0.027, non-significant). The interpretable within-Continental comparison, RUB vs TGH, holding route type constant, yields (waiting for ordered model results), confirming RUB's advantage within the Continental fleet. Yet TGH consistently outranks RUB in SHAP interaction importance for Promoters and Passives. The resolution is that these two metrics measure different things. RUB is a flat baseline shift relative to TGH within the Continental fleet, uniform NPS uplift across all passengers regardless of journey conditions, captured well by a main-effects model. TGH's effect is conditional: it interacts with delay, disruption, and staff perception to determine who becomes a Promoter, and that conditional relationship is what SHAP interaction values pick up. To verify this was not due to sample size, with RUB's smaller share of the data (3.5% of rows), the full interaction rankings across all six groups (2 feature sets $\times$ 3 classes) were checked. TGH's best rank per group for Promoters and Passives ranges from #1 to #6; its best rank for Detractors is #73 at best. RUB's best rank in any group is #35, and it typically sits between #100 and #350. TGH's best interaction magnitude also exceeds RUB's by a factor of 3.8 to 31 times across groups, too large and too consistent to be explained by sampling dilution alone. The operational implication is that TGH is a ceiling-side lever: improving conditions on TGH-operated services is more likely to convert Passive passengers into Promoters than to rescue passengers who have already become Detractors. RUB's value lies in its reliable baseline lift.

Interaction Effects

Pairwise SHAP interaction values identify feature pairs where the joint contribution differs from what each feature contributes individually. The top interactions are reported in Table Z.

Rank	Feature 1	Feature 2	Interaction (operational)	Interaction (price+staff)
1	arrival_delay_minute	had_disruption	0.0438	0.0168
2	arrival_delay_minute	early_journey_delay_minute	0.0318	0.0087
3	arrival_delay_minute	staff_composite	—	0.0244
4	metadata_price	staff_composite	—	0.0225
5	delay_gt30	early_journey_delay_minute	0.0072	—

Table Z. Top pairwise SHAP interactions (regressor)

$\text{arrival_delay} \times \text{had_disruption}$ is the strongest interaction in both feature sets. A journey combining a recorded disruption event with a high arrival delay incurs a larger NPS penalty than either factor alone would predict. The most plausible reading is that disruption announcements amplify the experienced cost of delay.. The interaction shrinks from 0.044 to 0.017 once staff and price are included, which suggests that part of the compounding effect runs through staff perception of the disruption rather than through the delay itself.

$\text{arrival_delay} \times \text{early_journey_delay}$ confirms the dual-delay compounding hypothesis (H7). The interaction value of 0.032 in the operational feature set is not trivially small, and it persists at 0.009 after

adding staff and price, so it is not simply a proxy for some third variable. The shape of the interaction matters too: dependence plots show the compounding concentrated in the 25–50 minute range for early journey delay. Beyond roughly 50 minutes of early journey delay, additional early journey delay does not materially worsen NPS further. Arrival delay shows no such plateau and keeps declining monotonically out to 300+ minutes.

When staff_composite is included, it produces interactions with both arrival_delay (0.024) and metadata_price (0.023) that rank second and fourth overall. This is not merely because staff is a strong main-effect predictor. The dependence plots show that low staff scores amplify the NPS penalty from both delay and high price, while high staff scores partially offset both. Staff perception is therefore a moderator of operational and price effects, not only an independent driver in its own right.

Finally, the SHAP dependence plots corroborated thresholds identified through decision trees. The zero-crossing for arrival_delay_minute in the operational-only dependence plot falls at approximately 15–20 minutes, consistent with the classifier tree's 19.5-minute threshold. The zero-crossing for metadata_price falls at roughly £90–100, consistent with the £94.50 split from the full tree.

Ordered Logit

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Describe and design the artifact (relevance cycle)

The artifact is a single interactive dashboard that brings the project's separate analytical threads, network structure, service-quality drivers, disruption thresholds, and the end-to-end NPS pipeline, into one tool Eurostar can actually use, rather than five disconnected notebooks each needing its own explanation. That follows directly from the gap identified at the start of this project: Eurostar already tracks operational KPIs and separately collects NPS, but nothing currently quantifies the link between the two or tells decision-makers where to act first. The dashboard's job is to close that gap for a non-technical, operationally-focused audience, not for a data scientist reviewing the underlying models.

That target audience shapes the layout. The entry point is a single summary screen: current NPS against Eurostar's 38-point target, the model's predicted NPS under the recommended operating point, and the single most influential lever, all readable in the time it takes to glance at a phone. Beneath that sits one tab per analytical workstream. Findings are written as short, plain-language statements rather than raw coefficients or p-values, with the underlying statistics, confidence intervals, model-fit indices, bootstrap stability percentages, kept visible but secondary, on hover or in a smaller note rather than the headline.

Accuracy is preserved by design. Every chart type in the mock-up is a direct stand-in for an output this project already produces elsewhere, SHAP rankings, ordered-logit coefficients with confidence ranges, threshold tables, goal-programming targets, so moving from placeholder to live numbers is a data swap, not a redesign. One colour convention, green for feasible or favourable, red for below threshold, amber for borderline, repeats across every tab, so the audience learns it once rather than relearning it per teammate's section.

Make a mock-up of your artifact (design cycle) (do we include images here?)

The first iteration was four separate dashboards, one per teammate's subquestion: a network-centrality view flagging which stations matter most to NPS, a service-quality view breaking NPS down by OLS and XGBoost coefficients across seven onboard dimensions, an end-to-end pipeline tying operational KPIs through perceived service quality to NPS via Bayesian-network discovery, path analysis, and an optimisation stage, and a delay-focused simulator with an interactive lever for testing fleet and turnaround scenarios. Each used real, if early-stage, numbers, but was built independently, with no shared visual language between them.

The purpose was to introduce the structure of the artifact to the Eurostar team, in addition to present the insights gathered until that point. The artifact included structured graphs, value placards, text (however, quite minimal at this point).

Evaluation

Review literature on evaluation metrics (rigor cycle)

Evaluation is aligned with the role of each method in the analytical pipeline. The preliminary decision tree, XGBoost, SHAP analysis and ordered logit do not serve the same purpose, so they should not be judged using one common measure. The decision tree provides an interpretable first view of possible thresholds and conditional relationships; XGBoost tests whether these patterns improve prediction on unseen services; SHAP examines the direction, shape and stability of the fitted relationships; and ordered logit quantifies the shortlisted thresholds while respecting the ordered nature of the recommendation outcome (Li et al., 2020; Monsuur et al., 2021).

The preliminary decision trees were evaluated using test-set (R^2) and root mean squared error for the 0–10 recommendation model, and ROC-AUC for the separate detractor, passive and promoter classifications. This combination distinguishes the amount of variation explained from the size of the prediction errors and assesses how well each NPS category can be separated across classification thresholds. The tree-depth comparison was used as an exploratory complexity check rather than formal optimisation, because the purpose of this stage was to identify readable candidate regions and interactions, not to select the final predictive model. This is consistent with the broader use of tree-based methods to reveal piecewise nonlinear and conditional satisfaction patterns before more flexible modelling (Gao et al., 2021).

XGBoost is the main predictive model and is therefore evaluated more strictly using service-grouped cross-validation. For regression, (R^2) and root mean squared error are reported; for multiclass NPS prediction, overall accuracy is considered alongside class-specific one-versus-rest AUC. Accuracy provides an accessible overall summary, but category-level measures are needed because aggregate performance can conceal whether detractors, passives and promoters are identified equally well. Studies comparing satisfaction and recommendation models similarly use multiple metrics and cross-validation rather than relying on a single accuracy result (Alanazi et al., 2024; Ruiz et al., 2024). Mean performance and variation across folds are reported because a model that performs strongly in one split but inconsistently across services would be less credible for Eurostar's operational use.

Training performance is also compared with grouped test performance. A large training–test gap is interpreted as possible overfitting, particularly where greater tree depth improves the training fit without producing comparable gains on unseen services. The preferred XGBoost configuration is therefore selected by balancing average predictive performance, fold-to-fold stability and generalisation rather than by choosing the highest isolated score. Early stopping provides an additional control by ending training when validation performance no longer improves, instead of fixing an unnecessarily large number of boosting rounds.

SHAP is not evaluated as a separate predictive model. Its value depends first on the performance and stability of the underlying XGBoost model. SHAP dependence plots are then assessed according to whether contribution patterns are coherent, sufficiently supported across the feature range and reproducible across grouped samples. Li et al. (2020) demonstrate that SHAP can be used to examine a proposed cut-point without assuming that a threshold must exist, while SHAP interaction values provide a way to isolate joint contributions beyond the corresponding main effects. Candidate thresholds should therefore be evaluated through their detection frequency, location across grouped resamples and uncertainty range, rather than from the visual appearance of one dependence plot. The full and lean datasets are also compared to determine whether the same patterns remain when variables with weaker operational coverage are removed.

The final ordered-logit model serves a confirmatory rather than exploratory purpose. Threshold and interaction variables supported by XGBoost and SHAP are entered explicitly, and their estimated coefficients, uncertainty intervals and statistical evidence are examined. Ordered logit is appropriate because the recommendation outcome has a meaningful ranking, while the distance between adjacent scores cannot be assumed to be exactly equal. Monsuur et al. (2021) provide the closest rail precedent by using ordered logit to quantify how objective delay conditions were associated with ordered passenger satisfaction. For Eurostar, a threshold is therefore considered suitable for prioritisation only when it is supported by grouped predictive performance, stable SHAP evidence and an ordered-logit estimate with an interpretable confidence range.

Design artifact MVP (design cycle)

The mock-up tested layout and tone; this step scopes down to what actually delivers, using the same MoSCoW logic already set out for the project as a whole. Must-have is the consolidated multi-tab structure itself: one general-insights tab plus one tab per teammate's subquestion. Within each tab, along with lines of must-have, the plain-language finding statements and the underlying graphs and charts that supports it, are carried over unchanged from the mock-up. It is ensured that the dashboard is consistent with one shared visual and colour convention, since it is what makes five different analyses feel like one tool rather than five.

To enhance UI, the dashboard covers the interactive layer beyond static charts: hover detail on the statistics behind each finding, and the scenario lever already prototyped on the delay-focused dashboard, extended to the tabs where a what-if control adds real decision value rather than just polish. Out of scope for this MVP is a live connection to Eurostar's own data systems. The deliverable is a working prototype that proves the design and the underlying analysis, populated by re-running the export-and-refresh process by hand, not a maintained production tool with automatic data feeds. That distinction matters for how the

artifact should be judged: as a demonstration of what a live version would do, not as that live version itself.

Evaluate with user feedback artifact utility and effectiveness; (relevant cycle)

Feedback on that round was clear on both a strength and a real problem. The depth and range of operational factors covered, and the visual structure showing how those factors connect to satisfaction and NPS, came across well. However, several charts were flagged as too technical to read without an operations background, and the dashboards fell short on translating findings into something a manager could act on directly. Further details regarding more practical insights and recommendations were advised to be added.

Update artifact based on feedback and evaluation (design cycle)

The artifact fixed for update, the interactive HTML version with simulated data, is the response to that feedback. It keeps the same tab logic, a summary screen plus one tab per analytical layer, but standardises the visual language across all of it: one dark theme, one colour convention, plain-language finding cards above each chart instead of dropping straight into statistics, and hover tooltips so detail is available without cluttering the default view. Every value in it is explicitly marked as simulated, since the point of this pass is to test the layout and the explanatory language before any real numbers go in. The order of insights shown is also ensured to be adjusted to smooth experience - a summary at first (preanalysis), followed by

Limitations & Further Research

Appendix - Artifact guidelines

Statement on work:

All the work (presented so far) is group work except the literature review for the modelling section, the subquestion introduction, which is my own individual work.

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